

# NIKHIL TULSANI

Senior QA Engineer | Transitioning to AI Engineer | Test Automation • Agentic AI • AI-Augmented QA

nikhil.tulsani1@gmail.com | +919574711771 | Ahmedabad, India

Portfolio: [nikhiltulsani1.github.io](https://github.com/nikhiltulsani1) | [linkedin.com/in/nikhil-tulsani-3a2575126](https://linkedin.com/in/nikhil-tulsani-3a2575126) | [github.com/nikhiltulsani1](https://github.com/nikhiltulsani1)

## PROFESSIONAL SUMMARY

I'm a Senior QA Engineer with 8 years of hands-on experience in manual testing, automation testing, and end-to-end quality ownership across SaaS, BFSI, and Healthcare domains. I'm currently transitioning into AI Engineering — on a live healthcare account, I built a production AI-powered platform using OpenAI Codex (database schema design through full AWS deployment), bridging directly from AI-assisted test automation with GitHub Copilot. I now serve as Team Lead for AI-augmented development on the same account. I work with agentic AI architecture (Model Context Protocol, LangGraph, LangChain, RAG) and have shipped personal AI projects including a hybrid-retrieval RAG system with guardrails and full observability, an eval-gated AI test-generation system, and two multi-agent hackathon pipelines. I architect automation frameworks using Selenium, Playwright, Appium, and Katalon, consistently reducing regression effort and accelerating release cycles.

## CORE COMPETENCIES

Test Automation • Manual Testing • API Testing • Backend Testing • Load Testing • Functional Testing • Regression Testing • Integration Testing • Test Planning • Defect Management • BDD / TDD • Agile / Scrum • CI/CD • SDLC • Team Leadership • Agentic AI • AI-Augmented Testing • Production AI Delivery

## TECHNICAL SKILLS

- **Automation Tools:** Selenium, Appium, Playwright, Katalon, REST-Assured, SoapUI
- **AI & Agentic Tools:** OpenAI Codex, GitHub Copilot, Claude AI, Claude Code, LangChain, LangGraph, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), hybrid search (BM25 + dense), RAGAS evaluation
- **Programming:** Core Java, Python
- **API & Backend:** REST API Testing, SoapUI, Database Testing, Message Broker Testing, PostgreSQL, OpenSearch
- **Performance Testing:** Apache JMeter (Load Testing, Performance Benchmarking)
- **Frameworks:** Cucumber, Gherkin (BDD), Data-Driven Framework, Page Object Model (POM)
- **DevOps & Cloud:** Jenkins, GitHub, Git, Azure DevOps, CI/CD Pipeline Integration, AWS (ECS, API Gateway, S3, Secrets Manager), Docker, n8n (Workflow Orchestration), OpenTelemetry
- **Platforms:** Web, Mobile (Android/iOS), API, SaaS, Healthcare, BFSI Applications

## PROFESSIONAL EXPERIENCE

### Senior Engineer – Quality Assurance

Sep 2022 – Present

Hexaware Technologies | Ahmedabad, India

- Owned end-to-end test automation on a live healthcare account spanning database validation, message-queue pipelines, API layer, and UI — including coverage across dependent downstream applications.
- Used GitHub Copilot for AI-assisted test automation, integrating AI-tooling into the SDLC for faster script generation and review cycles — this experience became the bridge into a Team Lead role for AI-augmented development on the same account, since April 2026.
- Designed the database schema and built a production AI-powered platform using OpenAI Codex, deployed on AWS (ECS, API Gateway, S3/CDN, Secrets Manager) with CI/CD pipeline integration via Azure DevOps. Continue to actively lead this effort on the platform.
- Designed and implemented automation frameworks using Selenium, Playwright, and Katalon, reducing manual regression testing effort by approximately 60% and improving test cycle speed.
- Conducted defect triage and root-cause analysis sessions that improved defect resolution turnaround time by 30%, directly increasing sprint velocity.
- Mentored and coached junior QA engineers; championed a quality-first engineering culture within Agile Scrum squads.

### Senior QA Engineer

Jul 2021 – Sep 2022

Maruti Techlabs Pvt. Ltd. | Ahmedabad, India

- Owned end-to-end quality assurance for a flagship SaaS communications platform, including white-label partner deployments across multiple enterprise client environments.

- Built and maintained UI and API automation test suites using Selenium and REST-Assured, achieving over 80% automated test coverage.
- Led a QA team of 4 engineers — facilitated daily Scrum meetings, reviewed test cases, tracked sprint velocity, and resolved team blockers.

## QA Engineer

Jul 2020 – Jul 2021

Maruti Techlabs Pvt. Ltd. | Ahmedabad, India

- Architected UI and API automation frameworks from scratch using Selenium and REST-Assured, establishing the team's baseline test infrastructure.
- Executed functional, integration, regression, and sanity testing for a multi-tenant SaaS platform and its white-label derivatives.
- Performed backend testing covering message brokers, schedulers, data stores, and REST APIs; maintained real-time error-log monitoring dashboards.
- Conducted load testing and API performance benchmarking using JMeter, identifying performance bottlenecks that informed the product's scalability roadmap.

## Associate Quality Analyst

Jun 2018 – Jul 2020

Streebo Solutions Pvt. Ltd. | Ahmedabad, India

- Delivered functional and manual testing across multiple BFSI client projects spanning Insurance, Banking, and internal enterprise applications.
- Executed load testing using Apache JMeter for a banking application's backend services; produced detailed performance reports presented to executive stakeholders.
- Built a mobile automation testing framework using Appium with BDD (Cucumber/Gherkin) and data-driven methodology for the company's flagship mobile product.

## PERSONAL PROJECTS

### Calibrated RAG — Hybrid RAG System with Guardrails

- Built a RAG system for querying research papers, designed to abstain rather than guess when source material doesn't support an answer. Hybrid retrieval (BM25 + dense, RRF-fused), a self-correcting retry loop, and 4 guardrails (fail-open/fail-closed by design) gate every response. 0.98 RAGAS faithfulness, validated against the published BEIR SciFact benchmark. Fully traced with OpenTelemetry; 180 automated tests.

### DriftPin — AI Test Generation System

- Multi-agent CLI converting PRDs into test cases with full requirement traceability — ambiguities flagged to an ASSUMPTIONS.md instead of invented coverage. Release 1 shipped eval-gated: 85% recall on golden PRD datasets, adversarial spec-defect detection against 9 planted defects, 252 passing tests (ruff/mypy clean), 4 LLM providers verified.

### PRD-to-Sprint Negotiator

- Built a 5-agent pipeline that converts a Product Requirements Document into a sprint backlog with Model Context Protocol (MCP)-compatible output for Jira, Azure DevOps, and Linear. Microsoft Agents League Hackathon 2026, Reasoning Agents track.

### Project Kickstart Agent

- Built a CLI tool that generates a production-ready GitHub repository from a plain-English description in under 60 seconds, supporting five programming languages with CI pipelines for each. Microsoft Agents League Hackathon 2026, Creative Apps track.

## EDUCATION

MCA — L.D. College of Engineering, Ahmedabad | CGPA 8.28/10

2018

BCA — Shree Swaminarayan College of Computer Science, Bhavnagar | CPI 8.18/10

2015

## CERTIFICATIONS

- **Claude Certified Associate – Foundations:** Anthropic (Pearson VUE, proctored) — Passed 884/1000. Issued Aug 2026, valid through Aug 2027.
- **Claude Partner Badge – Claude Code:** Anthropic (Credly) — 8 courses + scenario-based assessment: deployment, configuration, security, hooks, telemetry.

## ACHIEVEMENTS

- Elite Engineer Recognition Award — Hexaware Technologies, May 2026, for delivering a production AI-powered platform on a live healthcare account.

**Languages:** English (Professional), Hindi (Fluent), Gujarati (Native), Sindhi (Native)